GIOVANNI A. CAROSSO, PhD

Bioengineer | Founding Head of Research | Therapeutics Platforms

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SUMMARY

Bioengineer and startup operator specializing in discovery and translation of novel therapeutic modalities. Scientific lead, first hire, and lead inventor at General Control; set R&D strategy and built technology platform advancing from concept to *in vivo* validation within first year; owned protein and peptide engineering campaigns behind a multi-program pipeline of epigenetic editors, enabling seed financing and pharma-partner milestone. Previously founding scientist at Epicrispr Biotechnologies; co-developed GEMS modality platform behind EPI-321 (clinical-stage FSHD) and EPI-331 (DMD); lead inventor on engineered epigenetic activation technologies. Deep domain expertise in genetic engineering, epigenetic editing, CRISPR activation/inhibition, functional genomics, large protein language models, mRNA-LNP delivery, preclinical diligence, competitive target diligence (cardiometabolic, muscular, longevity), and investor/pharma diligence.

TECHNICAL SKILLS

Platform architecture & operations: high-throughput CRISPRa/CRISPRi screens, proteome-scale library assembly, scalable plasmid topology, mRNA engineering, transcriptomics, NGS, Python (CLI tooling: NumPy, pandas, Biopython), R (tidyverse, ggplot2, Quarto), Git/GitHub, agentic toolchains.
ML-guided protein engineering: LPLM-based sequence–function modeling (ESM-2 embeddings, CNNs/XGBoost), evolutionary sequence optimization (EMCS), generative binder design (BoltzGen), and structure-based candidate evaluation (Boltz-2).
Preclinical strategy: modality selection, *in vivo* efficacy/safety, dose-response modeling, IND-enabling program design.

EXPERIENCE

General Control Jul 2024 – Jul 2026

Founding Head of Research (first hire)

- Delivered *in vivo* target engagement within 12 months from program ideation and positive *in vivo* readouts across six of seven programs inside 18 months, by designing and deploying rapid engineering campaigns of novel epi-editors against industry benchmarks; established single-dose, multiplexed targeting as a modality differentiator, delivering >90% knockdown of three targets in parallel.
- Accelerated preclinical hit-translation cycles from 16 to 3 weeks by devising an IVT-compatible mammalian expression vector that allows high-throughput screening elements to proceed directly into the mRNA synthesis pipeline for animal dosing.
- Optimized editor compactness, potency, and novelty profiles with a model-guided search strategy over peptide sequence space that generates diverse candidates, ranks by predicted function and structural confidence, and concentrates experimental resources on high-probability sequences. 155 novel editors outperformed benchmarks across diverse genetic targets.
- Supported company financing and strategic partnership with Novo Nordisk, connecting GC's proprietary platform development with pharma-partner objectives; shaped scientific IP and FTO strategy as lead inventor with counsel through PCT filings and evaluation of competitive patent landscapes.
- Co-recruited the team as GC grew to five FTEs and coordinated execution across internal scientists, computational contractors, CROs, and advisers, with responsibility for scientific priorities, lab instrumentation, preclinical milestones, and resource-allocation decisions.

Epicrispr Biotechnologies Jan 2021 – Jul 2024

Senior Scientist, Technology Development (previously Scientist II, Scientist)

- Co-developed GEMS modality platform by introducing a compact Cas chassis and engineering proprietary epi-editor domains behind EPI-321, now in first-in-human FSHD trial (NCT06907875), and EPI-331 for durable activation.
- Led discovery and engineering of hypercompact activator, sized 64–98 residues, that sustained five-week *LDLR* activation via single-dose mRNA-LNP *in vivo*; first author on external preprint article.
- Increased pooled-screen editor hit rates from <1% to 8%; generated sequence-function data and training set for collaborative ML-guided protein engineering. *NeurIPS* (2023).

University of California, San Francisco (UCSF) Jan 2020 – Dec 2020

Postdoctoral Scholar, Neurological Surgery; Lab of Daniel Lim, MD, PhD

- Integrated Perturb-seq, PLAC-seq, ChIP-seq and Hi-C into a machine-learning classifier over genome-scale dual CRISPRi screens spanning 18,905 coding and 10,678 lncRNA loci, with the Weissman and Lim Labs. *Cell Genomics* (2022).

OPEN-SOURCE PORTFOLIO

SAROS — clinical & biopharma research dashboard | GitHub

– Integrates clinical-trial, regulatory, and securities data through API ingestion and entity resolution (JavaScript, Python); automated data refreshes, testing, and deployment (Make, launchd, Cloudflare Pages).

Pileup — Agent execution visualizer | GitHub

– Live visualization of tool calls, subagents, errors, and context usage through incremental JSONL parsing and server-sent events (Python, JavaScript/Canvas); dependency-free local execution with optional content redaction.

EDUCATION

Johns Hopkins University School of Medicine 2019

PhD, Human Genetics; Lab of Hans T. Bjornsson, MD, PhD

– Discovered molecular drivers in pediatric disorders of chromatin machinery, novel roles for KMT2D in oxygen sensing during neurodevelopment. Dissertation, *JCI Insight* (2019).

– Secured NASA funding award for astronaut epigenetics in the ISS Twins Study Consortium (Andy Feinberg Lab); co-developed first protocols for in-flight epigenomic sampling aboard ISS and experiments for astronauts Scott and Mark Kelly.

University of California, Santa Barbara 2010

BS, Biopsychology

– Discovery of *POLR3A* mutations causing pediatric leukodystrophy. *AJHG* (2011).

– Elucidated mechanisms of medial prefrontal cortex dysfunction in cocaine abuse. *Addiction Biology* (2012).

ADVISORY, PUBLIC ENGAGEMENT

STEM diplomacy: Led a U.S. Department of Defense-funded neuroscience workshop with *Clubes de Ciencia Bolivia*; taught in Spanish and published on science diplomacy.

Mindfire Global — Neurobiology consultancy; advised on neurodevelopment and human-centric AI collaboration.
Languages: Italian, French, Spanish, English.

PATENT APPLICATIONS

US 2024/0216482 A1	—	Systems and methods for regulating aberrant gene expressions.
US 2024/0254659 A1	—	Systems and methods for regulating target genes.
WO 2023/172995 A1	—	Systems and methods for genetic modulation to treat ocular diseases.
US 2025/0388634 A1	—	Engineered gene effectors, compositions, and methods of use thereof.
US 2026/0085300 A1	—	Systems and compositions for fusion polypeptides and methods of use thereof.
WO 2024/238661 A2	—	Systems and methods for regulating target genes.
PCT/US26/24038	—	Compositions and Methods for Epigenome Editing.
PCT/US26/24051	—	Systems and Methods for Upregulation of Exerkines.

PUBLICATIONS

Carosso GA et al. (2024). Discovery of hypercompact epigenetic modulators for persistent CRISPR-mediated gene activation. *bioRxiv*.
Wu D et al. (2022). Dual genome-wide coding and lncRNA screens in neural induction of induced pluripotent stem cells. *Cell Genomics*, 2(11), 100177.
Zheng SC et al. (2022). Universal prediction of cell-cycle position using transfer learning. *Genome Biology*, 23(1), 41.
Breevoort A et al. (2020). High-altitude populations need special considerations for COVID-19. *Nature Communications*, 11, 3280.
Carosso GA et al. (2019). Precocious neuronal differentiation and disrupted oxygen responses in Kabuki syndrome. *JCI Insight*, 4(20), e129375.
Carosso GA et al. (2019). Scientists as non-state actors of public diplomacy. *Nature Human Behaviour*, 3(11), 1129–1130.
Carosso GA et al. (2019). Developing brains, developing nations: can scientists be effective non-state diplomats? *Frontiers in Education*, 4, 95.
Ferreira LMR et al. (2019). Effective participatory science education in a diverse Latin American population. *Palgrave Communications*, 5(1), 11.
Benjamin JS et al. (2016). A ketogenic diet rescues hippocampal memory defects in a mouse model of Kabuki syndrome. *PNAS*, 114(1), 125–130.
Vanderver A et al. (2013). More than hypomyelination in Pol-III disorder. *Journal of Neuropathology & Experimental Neurology*, 72(1), 67–75.
Ben-Shahar OM et al. (2012). Extended access to cocaine self-administration results in reduced glutamate function within the medial prefrontal cortex. *Addiction Biology*, 17(4), 746–757.
Bernard G et al. (2011). Mutations of POLR3A encoding a catalytic subunit of RNA polymerase Pol III cause a recessive hypomyelinating leukodystrophy. *American Journal of Human Genetics*, 89(3), 415–423.